



Quanto derivatives in local volatility models

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1 Motivations

2 Approximations based on the Proxy Expansion technique

- Proxy Expansion toolkit
- Application to local volatility calibration
- Quanto pricing formula

3 Study case

Outline

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Motivations

- Focus on quanto forward **marking** and quanto derivs pricing
- The Black-Scholes classic quanto drift approximation may have been used in the past to mark and risk manage the quanto covariance
- The 2008 crisis as illustrated how flawed that was esp. for exotics, and since then practitioners have been addressing the issue of unifying quanto 'vanillas' and quanto exotics pricing and risk management
- We will describe in this presentation an approximation technique to achieve fast pricing of quanto forward and options under local volatility
- References include (Jäckel, 2009) and (Jäckel, 2010)

Quanto options 101

- We assume a local volatility diffusion dynamics for the equity and FX under the domestic measure

$$\begin{aligned} dS_t/S_t &= (\mu(t) - \rho\sigma_S\sigma_X)dt + \sigma_S dW_t \\ dX_t/X_t &= (r_d - r_f)dt + \sigma_X dW'_t \\ d\langle W_t, W'_t \rangle &= \rho dt \end{aligned} \quad (1)$$

where we denote with X the value of one unit foreign currency under the domestic economy. We will consider later NKY and JPYUSD.

- Guaranteed Exchange Rate options: a domestic investor receives an payout written on a foreign underlying at a guaranteed exchange rate. For a call, the forward price writes as

$$\mathbb{E}^d \left[\left(S_T^f - K \right)^+ \right] = \mathbb{E}^f \left[\frac{X_0}{X_T} \left(S_T^f - K \right)^+ \right]$$

Quanto options 101

- PnL of a short position of a quanto option V that is equity and FX delta-hedged:

$$\begin{aligned} d\text{PnL}(t) &= \frac{1}{2} \Gamma S^2 \left(\sigma_{S,\text{hedge}}^2 - \sigma_{S,\text{real}}^2 \right) dt \\ &+ \Delta S \left(\text{real quanto cov} - \text{hedge quanto cov} \right) dt \end{aligned}$$

- Equity-FX covariance affects valuation through the equity delta, which means that even the humble quanto forward is EQ/FX skew dependent

$$F_T^{\text{qto}} = F_T \mathbb{E}^d \left[e^{-\int_0^T \rho \sigma_X \sigma_S dt} \right]$$

- Question: if the quanto correlation is derived from market quotes of quanto forwards, what vol modeling assumption shall we use to imply the correlation?

Standard quanto drift approximation

- Historically, practitioners kept it super simple. (Jäckel, 2009) provides a good review of those standard pricing rules for quantos.
- **Quanto forward:** a standard pricing rule relies on a quanto drift derived in the Black-Scholes world

$$F'_T = F_T e^{-\rho \hat{\sigma}_S^{\text{atmf}} \hat{\sigma}_X^{\text{atmf}} T}$$

where we used the at-the-money-forward implied Black vols sampled at expiry T

- **Quanto options:** a widespread convention consists in using the Black-vol of the domestic option sampled at the same absolute strike

$$\mathbb{E}^d \left[\left(S_T^f - K \right)^+ \right] = \text{Black} \left(F'_T, K, \hat{\sigma}_{\text{eq}}(T, K), T \right)$$

Reconciling BS quanto drift approximation with exotics

- When it comes to pricing exotics, **local volatility** is a popular choice but using the BS implied quanto correlation will not replicate the marked quanto forward... how to solve this conundrum?
- Simplest answer is a **fudge factor** approach: in the exotic model, describe the quanto underlying with a standard one-factor model but with the deterministic forward curve F' and the domestic implied volatility surface, i.e. simulate quanto asset with

$$dS^f / S^f = \left(r^f - \underbrace{\rho \hat{\sigma}_S^{\text{atmf}} \hat{\sigma}_X^{\text{atmf}}}_{\text{BS quanto drift}} \right) dt + \sigma_{S^f}^{\text{loc}} dW$$

- Note that $\sigma_{S^f}^{\text{loc}}$ is derived from the modified forward curve and the foreign implied vol surface

Why would we need to do better?

- Despite its simplicity, using the Black-Scholes quanto methodology for exotics has a number of drawbacks
- **Wrong price and risks:** if we hedge with the natural underlying, premium does not reflect exposure to joint moves in volatilities and spots even though we are exposed to it through Δ . This is potentially even worse for exotics which have uncertain duration and variable Δ as spot moves.
- **FX skew risk is not captured**
- **Biais in the quanto options implied vol recipe:** the difference with say local vol may flip sign depending on correlation and the FX and EQ skews
- **The FX composite becomes arbitrageable:** when the derivative also depends on the FX (e.g. callable hybrid range accrual), the model's composite forward becomes arbitrageable

$$\mathbb{E}^d \left[S_T^f X_T \right] \neq \mathbb{E}^f \left[S_T^f \right] \mathbb{E}^d \left[X_T \right]$$

Solutions

- Solve the problem at the source: replace the BS marking model for the quanto forward and options with a local volatility one
- We need a fast and robust method to generate the quanto forward and option prices from the quanto correlation and the EQ and FX implied local volatility surfaces
- Numerical approach: we could use a 2-factor Kolmogorov-forward PDE to mark the quanto correl but that might be deemed too costly for a marking model

Solutions

- Alternatives based on closed-form formula can be found in the literature and include displaced diffusion in (Jäckel, 2009) or stoch vol (Jäckel, 2010)
- These approaches convey some insights but may not be applicable in a production environment
- We propose here a closed-form approximation that presents 2 advantages:
 1. derived under a local volatility assumption for both EQ and FX
 2. based on a time-dependent parameterisation of the local volatility that better captures the pronounced skews of EQ and FX implied vols

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Proxy Expansion

- We apply the Proxy Expansion technique pioneered by (Benhamou, Gobet, Miri, 2009) and based on approximating the sample value of a Proxy process by an expansion around a simpler one that is tractable, e.g. LN or Normal
- Successfully used for Heston and local volatility model in a single underlying framework. Applications to multi-underlying include (Hok, 2013) which models equity with a displaced diffusion and inflation with a LN assumption
- This work extends previous studies in 2 ways:
 1. derive the formula of the bivariate payoff where both processes are local volatility-based
 2. apply it to a parametric local volatility that better captures the term-structure of market smiles than CEV or displaced-diffusion

Proxy Expansion: mono-underlying case

- We model each process process by a general one-factor SDE

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 = x_0$$

- Introduce a family of processes indexed by η denoted with X_t^η :

$$dX_t^\eta = \eta (\mu(t, X_t^\eta) dt + \sigma(t, X_t^\eta) dW_t)$$

- Approximate the original process by a Taylor expansion around $\eta = 0$
- Let us denote with $X_{i,t}$ the derivative of the process with respect to the scaling factor η :

$$X_{i,t} = \left. \frac{\partial^i X_t^\eta}{\partial \eta^i} \right|_{\eta=0}$$

Proxy Expansion: mono-underlying case

- The derivative processes are described by the following system of coupled SDEs:

$$\begin{aligned}dX_{1,t} &= \mu_t dt + \sigma_t dW_t \\dX_{2,t} &= 2X_{1,t} (\mu'_t dt + \sigma'_t dW_t) \\dX_{3,t} &= 3X_{1,t}^2 (\mu''_t dt + \sigma''_t dW_t) + 2X_{2,t} (\mu'_t dt + \sigma'_t dW_t)\end{aligned}\quad (2)$$

where all the coefficients denote the values or derivatives wrt the space variable **frozen at the initial point** x_0 , e.g.

$$\mu'_t = \frac{\partial \mu(t, x)}{\partial x} \Big|_{x=x_0}$$

- We can rewrite X as a Taylor expansion around X_0 wrt to η :

$$X_T \approx X_0 + X_{1,T} + \frac{1}{2}X_{2,T} + \frac{1}{6}X_{3,T} \quad (3)$$

Proxy Expansion approximation: mono-underlying case

- Approximate the expectation of a payoff function $F(X_T)$ by a standard Taylor expansion of F around the normal Proxy process $X_{1,T}$
- We here omit the third order derivative process and also content ourselves with a first order Taylor approximation of F :

$$F(X_T) \approx F(X_0 + X_{1,T}) + F'(X_0 + X_{1,T}) \frac{1}{2}X_{2,T} \quad (4)$$

- The expectation of the first term is analytically tractable. The second term can be transformed into a weighted sum of the sensitivities of the payoff expectation wrt x_0 calculated with the Gaussian proxy process X_1

Proxy expansion approximation: mono-underlying case

- Following Gobet et al. notations, the expansion writes as:

$$\begin{aligned}\mathbb{E}[F(X_T)] &\approx \mathbb{E}[F(x_0 + X_{1,T})] \\ &\quad + C_{1,T} \frac{\partial \mathbb{E}[F(x_0 + X_{1,T})]}{\partial x_0} \\ &\quad + C_{2,T} \frac{\partial^2 \mathbb{E}[F(x_0 + X_{1,T})]}{\partial x_0^2} \\ &\quad + C_{3,T} \frac{\partial^3 \mathbb{E}[F(x_0 + X_{1,T})]}{\partial x_0^3}\end{aligned}\tag{5}$$

- The coefficients are defined as

$$\begin{aligned}C_{1,T} &= \omega(\mu, \mu') \\ C_{2,T} &= \omega(\sigma^2, \mu') + \omega(\mu, \sigma\sigma') \\ C_{3,T} &= \omega(\sigma^2, \sigma\sigma')\end{aligned}$$

- We denoted with ω the double integral:

$$\omega(f, g) := \int_0^T f(t) \int_t^T g(u) du dt$$

Proxy expansion approximation: practical remarks

- The technique can be extended to a bivariate payoff of 2 correlated LV processes
- In practice, we use a 3rd order expansion* in order to capture dependencies wrt 2nd order spatial derivatives. This is critical for real-life vol smile which exhibits a strong curvature wrt strike, e.g. in FX markets
- Assuming piecewise time-homogenous local volatilities, an efficient implementation of the coefficient computation scales linearly with the number of periods
- We will apply those results to:
 1. efficiently calibrate a time-dependent parametric volatility to a term-structure of smiles
 2. derive an expansion formula for a bivariate payoff under local volatilities

(*) Third order because the correction terms are in order 3 of the volatility coefficient amplitude

Local volatility calibration

- We use the local variance is quadratic in $\log S$ proposed in (Reghaï, 2007)

$$dS_t/S_t = \sigma(t, S_t) dW_t, \quad X_0 = x_0$$

where $\sigma^2(t, S) = v(t)^2 + 2\rho(t)v(t)\alpha(t) \log S + \alpha^2(t) \log^2 S$

- The time-homogenous case is tractable as the implied volatility is derived from a general short-term asymptotic approximation result found in (Gatheral, 2009):

$$\Sigma(T, K) = \Sigma_0(K) + \Sigma_1(K)T + \Sigma_2(K)T^2 + \mathcal{O}(T^3)$$

The zero-th order term and first order term write as:

$$\begin{aligned} \Sigma_0(K) &= \frac{s-k}{f(s)-f(k)}, \quad f(x) := \frac{\log(\rho v + \alpha x + \sigma(x))}{c} \\ \Sigma_1(K) &= \frac{\Sigma_0^3(K)}{(s-k)^2} \log \left(\frac{\sqrt{\sigma(s)\sigma(k)}}{\Sigma_0(K)} \right) \end{aligned}$$

where $s := \log S$ and $k := \log K$

Local volatility calibration

- Parsimonious and ease of interpretation of the parameters (v, α, ρ) : the zero-th order matches the SABR model with $\beta = 1$
- Calibration to SX5E as of end of January 2014 with $(v, \rho, \alpha) = (20.5\%, -65.9\%, 35.8\%)$



Figure: Fitting to Dec16 options

- We still need a fast calibration technique for the time-dependent version

Local volatility calibration

- For the purpose of calibrating a time-dependent local vol, we rewrite the process as:

$$dS = \varphi(t, S)dW_t \quad , \quad \varphi(t, S)^2 \equiv S^2 \sigma^2(t, S)$$

- Apply similar techniques to (Benhamou, Gobet, Miri, 2009): derive an expansion formula around a normal proxy and match leading order terms in both the time-dependent and time-homogenous models
- An expansion formula can be derived as:

$$\begin{aligned} \mathbf{E} [(S_T - K)^+] &\approx \Phi(S_0, T, K, \Sigma_T) + \omega(\sigma^2, \sigma\sigma') \frac{\partial^3 \Phi(S_0, T, K, \Sigma)}{\partial S_0^3} \\ &\quad + (w(\sigma^2, \sigma\sigma', \sigma\sigma') + w(\sigma^2, \sigma^2, \sigma\sigma'')) \frac{\partial^4 \Phi(S_0, T, K, \Sigma)}{\partial S_0^4} \end{aligned}$$

where Φ is the Bachelier pricing formula, and all the coefficients denote the normal local vol frozen at its initial point

Local volatility calibration

- Match leading order terms in both the time-dependent and time-homogenous models
- After defining $\beta := \rho\alpha$, straightforward calculations lead to the following formulas:

$$\begin{aligned} \hat{v}^2 &= \frac{1}{T} \int_0^T v^2 dt \\ \frac{\hat{v}^3 T^2}{2} \hat{\beta} &= \int_0^T v \beta \int_0^t v^2 ds dt \\ \frac{\hat{v}^4 T^3}{6} (\hat{\alpha}^2 + 3\hat{v}\hat{\beta}) &= \int_0^T (\alpha^2 - \beta^2 + 2v\beta) \int_0^t v^2 \int_0^s v^2 du ds dt \\ &\quad + \int_0^T v(\beta + v) \int_0^t v \beta \int_0^s v^2 du ds dt \end{aligned}$$

where the terminal params are denoted with the hat symbol.

Local volatility calibration: numerical examples

Fitted parameters for USDJPY as of end of April 2013

T	v	ρ	α
3M	11.46%	6.80%	100.31%
6M	10.79%	5.68%	84.77%
9M	10.21%	4.55%	73.81%
1Y	10.12%	5.37%	66.36%
1Y6M	10.60%	-3.42%	54.55%
2Y	10.49%	-4.33%	46.16%
3Y	11.29%	-9.83%	40.80%
4Y	11.85%	-12.69%	35.18%
5Y	12.00%	-17.85%	29.95%

Local volatility calibration: numerical examples

Fitted parameters for Nikkei as of end of April 2013

T	v	ρ	α
3M	23.85%	-21.97%	104.10%
6M	21.43%	-20.38%	74.68%
9M	19.32%	-20.34%	61.76%
1Y	18.90%	-20.45%	55.37%
1Y6M	19.07%	-20.46%	49.26%
2Y	18.90%	-20.30%	43.59%
3Y	19.95%	-25.73%	38.73%
4Y	21.04%	-23.83%	34.23%
5Y	22.31%	-22.46%	31.50%

Local volatility calibration: the Nikkei index

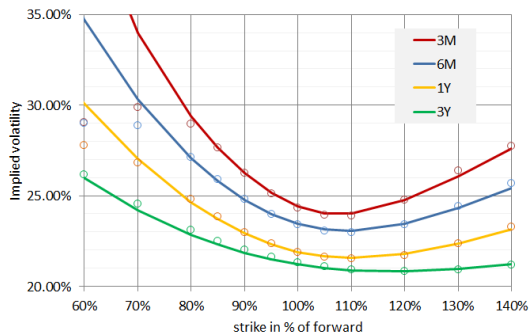


Figure: Formula vs. Monte-Carlo

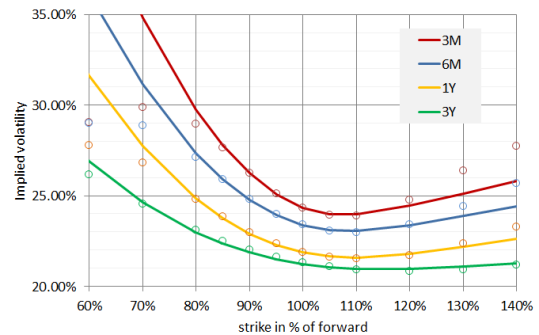


Figure: Formula vs. Market

Local volatility calibration: USDJPY 3M

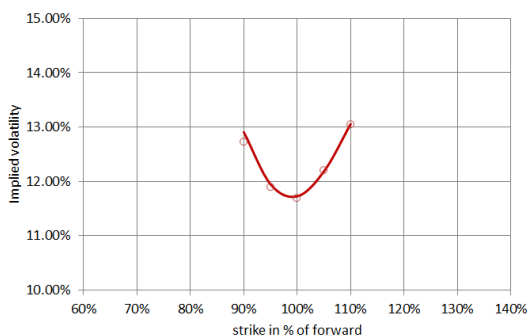


Figure: Formula vs. Monte-Carlo

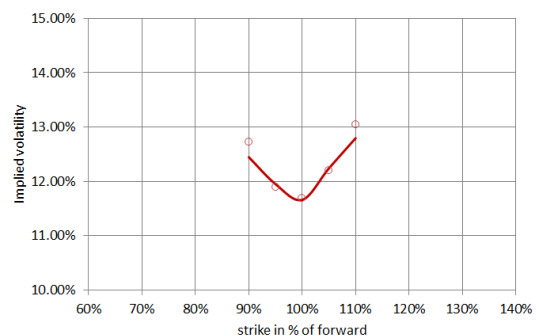


Figure: Formula vs. Market

Local volatility calibration: USDJPY 6M

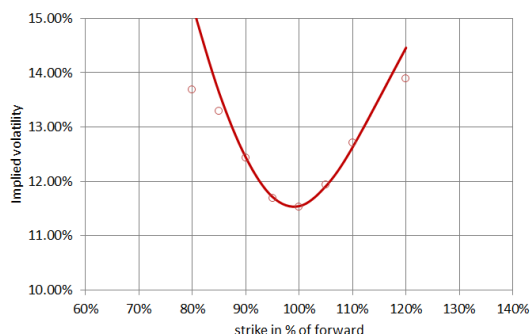


Figure: Formula vs. Monte-Carlo

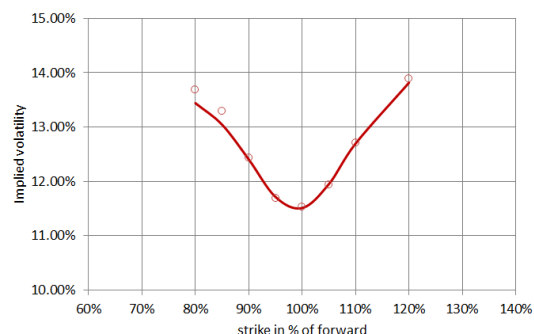


Figure: Formula vs. Market

Local volatility calibration: USDJPY 1Y

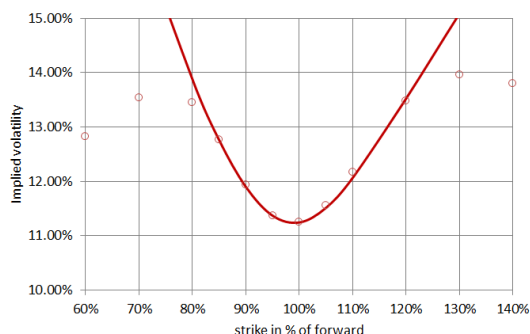


Figure: Formula vs. Monte-Carlo

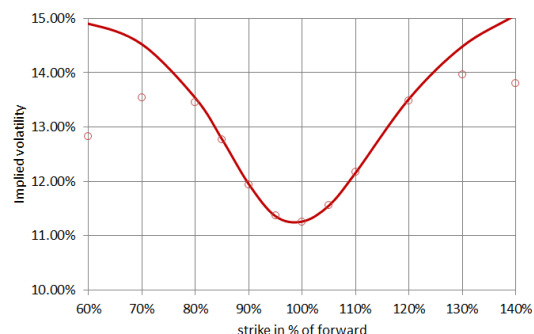


Figure: Formula vs. Market

Local volatility calibration: USDJPY 3Y

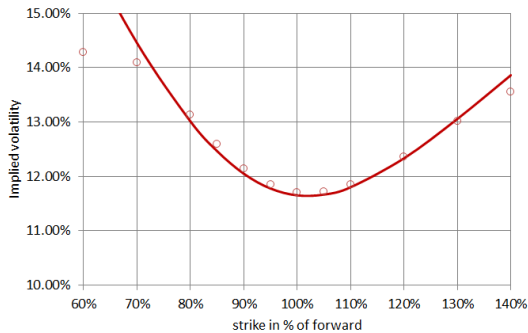


Figure: Formula vs. Monte-Carlo

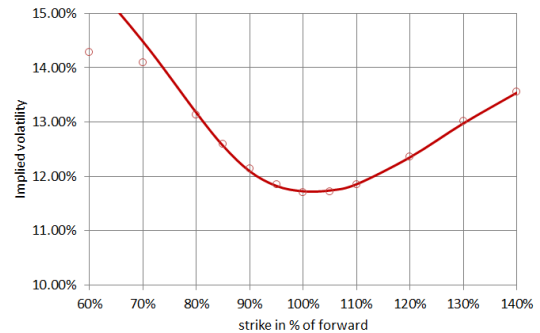


Figure: Formula vs. Market

Expansion formula for quantos

- Without loss of generality we now write the quanto payoff under the foreign measure as:

$$\mathbb{E} [\exp(X) \max(0, \exp(Y) - K)]$$

where we respectively denoted with X and Y the FX and equity process described by the SDEs:

$$\begin{aligned} dX_t &= -\sigma_X^2(t, X_t)/2 dt + \sigma_X(t, X_t) dW_t^X, \quad X_0 = 0 \\ dY_t &= -\sigma_Y^2(t, Y_t)/2 dt + \sigma_Y(t, Y_t) dW_t^Y, \quad Y_0 = 0 \\ d\langle W_t^X, W_t^Y \rangle &= \rho dt \end{aligned}$$

Notice that we now denote with X the value of 1 unit of domestic currency under the foreign economy

- We assume that the local variance are piecewise time-homogenous and defined by the $v - \alpha - \rho$ parameterisation introduced earlier

Expansion formula for quantos

- We approximate each process using the Proxy expansion technique et expand the payoff as:

$$F(X_T, Y_T) \approx F(X_{1,T}, Y_{1,T}) + \varepsilon_X + \varepsilon_Y + \varepsilon_{XX} + \varepsilon_{YY} + \varepsilon_{XY} \quad (6)$$

where the following notations are used:

$$\varepsilon_X := \frac{1}{2} X_{2,T} \frac{\partial F}{\partial x}(X_{1,T}, Y_{1,T})$$

$$\varepsilon_{XX} := \frac{1}{8} X_{2,T}^2 \frac{\partial^2 F}{\partial x^2}(X_{1,T}, Y_{1,T}) + \frac{1}{6} X_{3,T} \frac{\partial F}{\partial x}(X_{1,T}, Y_{1,T})$$

$$\varepsilon_Y := \frac{1}{2} Y_{2,T} \frac{\partial F}{\partial y}(X_{1,T}, Y_{1,T})$$

$$\varepsilon_{YY} := \frac{1}{8} Y_{2,T}^2 \frac{\partial^2 F}{\partial y^2}(X_{1,T}, Y_{1,T}) + \frac{1}{6} Y_{3,T} \frac{\partial F}{\partial y}(X_{1,T}, Y_{1,T})$$

$$\varepsilon_{XY} := \frac{1}{4} X_{2,T} Y_{2,T} \frac{\partial^2 F}{\partial x \partial y}(X_{1,T}, Y_{1,T})$$

The expectation of the first term is our standard BS valuation using ATM-F vols.

Expansion formula for quantos

- In practice, we have noticed that the contribution of ε_{XY} is negligible
- Expanding each the other terms still lead to the calculation of close to 90 coefficients involving up to 4 levels of nesting, albeit they are cheap to compute
- For the sake of concision, we here reproduce only the 2nd order terms, i.e. we approximate the expectation as:

$$\mathbb{E}[F(X_T, Y_T)] \approx \mathbb{E}[F(X_{1,T}, Y_{1,T})] + \varepsilon_X + \varepsilon_Y$$

where $F(x, y) = e^x (y - K)^+$

- Numerical examples will show the results obtained with the contribution of the 3rd order terms.

Expansion formula for quantos

- The expectation is approximated as a valuation in the BS world complemented by a series of correction terms

$$\begin{aligned}\mathbb{E}[F(X_T, Y_T)] \approx & \mathbb{E}[F(X_{1,T}, Y_{1,T})] + \frac{1}{2}C_1 G_{10} + \frac{1}{2}C_5 G_{01} \\ & - \frac{3}{2}C_1 G_{20} - \frac{3}{2}C_5 G_{02} \\ & - (\frac{1}{2}C_3 + C_2 + \frac{1}{2}C_7 + C_6) G_{11} \\ & + C_1 G_{30} + C_5 G_{03} \\ & + (C_2 + C_3 + C_8) G_{21} + (C_4 + C_7 + C_6) G_{12}\end{aligned}$$

$$\text{with } G_{ij} = \frac{\partial^{i+j} \mathbb{E}[F(x_0 + X_{1,T}, y_0 + Y_{1,T})]}{\partial x_0^i \partial y_0^j} \Big|_{x_0=0, y_0=0}$$

- The coefficients $(C_i)_{1 \leq i \leq 8}$ are defined as nested integrals:

$$\begin{aligned}C_1 &= \omega((\sigma_X)^2, \sigma_X \sigma'_X) & C_2 &= \omega(\rho \sigma_X \sigma_Y, \sigma_X \sigma'_X) \\ C_3 &= \omega((\sigma_X)^2, \rho \sigma'_X \sigma_Y) & C_4 &= \omega(\rho \sigma_X \sigma_Y, \rho \sigma'_X \sigma_Y) \\ C_5 &= \omega((\sigma_Y)^2, \sigma'_Y \sigma_Y) & C_6 &= \omega(\rho \sigma_X \sigma_Y, \sigma'_Y \sigma_Y) \\ C_7 &= \omega((\sigma_Y)^2, \rho \sigma'_Y \sigma_X) & C_8 &= \omega(\rho \sigma_X \sigma_Y, \rho \sigma'_Y \sigma_X)\end{aligned}$$

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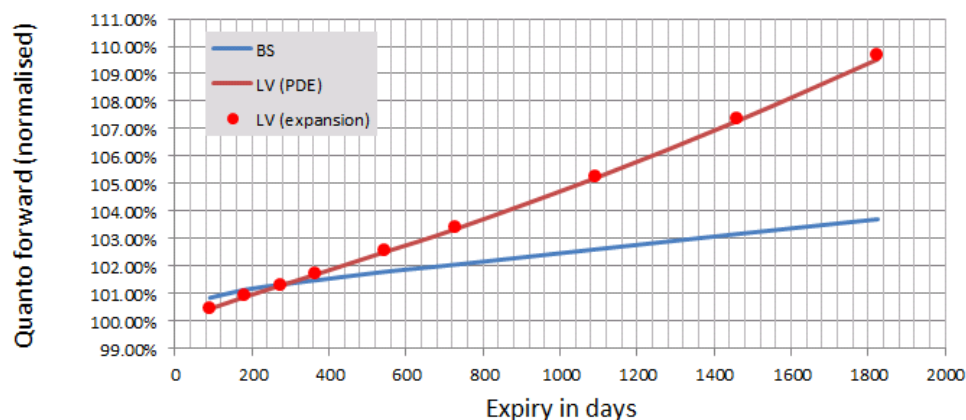
NKY/USD quanto forward

- We consider NKY quantoed into USD assuming 60% correlation between USDJPY and NKY using the same data as in the previous section, in particular we calibrate the local volatilities on a relatively dense set of expiries
- Our benchmark model is the PDE solution of a standard local volatility model for both EQ and FX. We also compare with the ATMF BS model.
- Vol model impact for the same input correlation well approximated by:

$$F_{LV}^{\text{quanto}} - F_{BS}^{\text{quanto}} \approx \int_0^T \rho \Delta S \left(\sigma_X^{LV} \sigma_S^{LV} - \sigma_X^{BS} \sigma_S^{BS} \right) dt$$

NKY/USD quanto forward

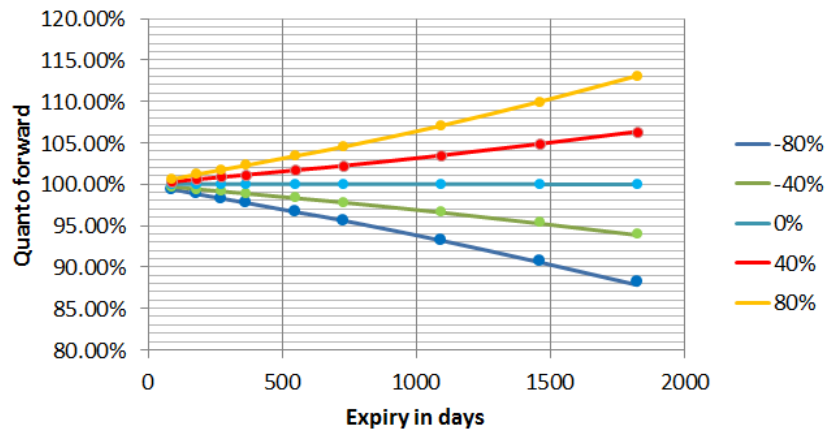
- The expansion formula accurately replicates the term-structure of quanto forward in LV



Note that we report here the normalised value, ie divided by the product of the EQ and FX forwards

NKY/USD quanto forward

- Consistency is achieved across a wide range of quanto correlation



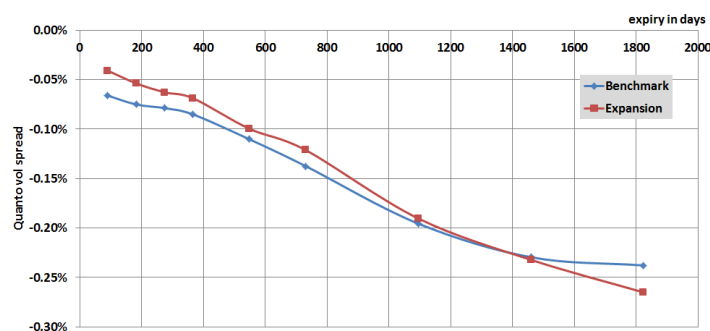
- This example illustrates that the expansion formula can be used as a marking model for implied quanto correlation term-structure

NKY/USD quanto options

- For options, we present the results in terms of the vol spread we need to add on top of the natural implied volatilities:

$$\Delta\hat{\sigma} := \hat{\sigma}_{LV}^{\text{quanto}} - \hat{\sigma}_{\text{natural}}$$

- Results for the quanto options struck ATMF (in moneyness of Nikkei forward)



Increasing EQ skew

- PnL tracking error of a delta-hedged long position omitting gamma terms

$$\text{PnL}(0, T) = \int_0^T \Delta(t, S) (\rho_{\text{real}} \sigma_{S, \text{real}} \sigma_{X, \text{real}} - \rho_{\text{theo}} \sigma_{S, \text{theo}} \sigma_{X, \text{theo}})$$

where X denotes USDJPY and $\rho := \text{corr}(\text{USDJPY}, \text{NKY})$

- If $\rho := \text{corr}(\text{USDJPY}, \text{NKY}) > 0$, steeper EQ skew will $\downarrow$ ATMF quanto implied vol
- If $\rho := \text{corr}(\text{USDJPY}, \text{NKY}) < 0$, steeper EQ skew will $\uparrow$ ATMF quanto implied vol

Increasing EQ skew

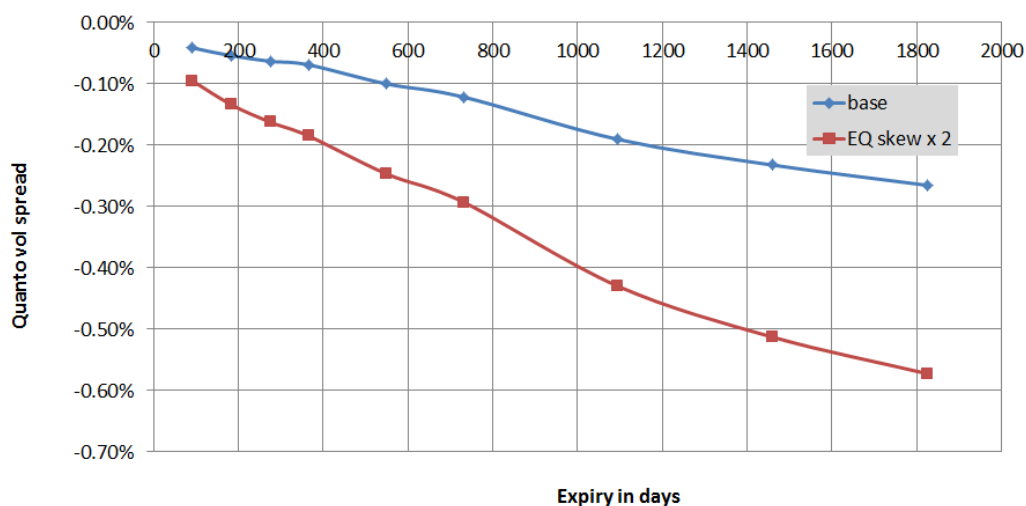


Figure: Effect of EQ skew x2 for $\rho > 0$

Increasing EQ skew

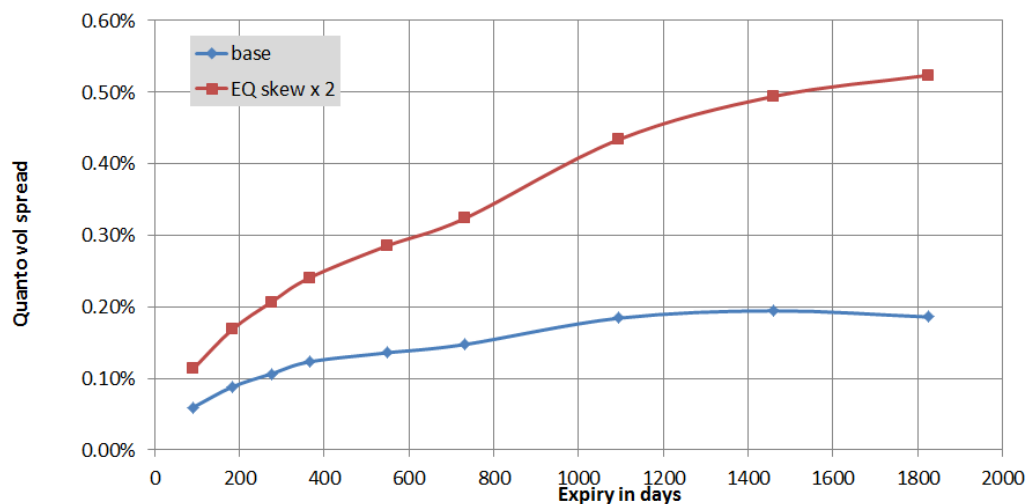


Figure: Effect of EQ skew x2 for $\rho < 0$

Conclusion

- We illustrated that the valuation of quanto derivs is sensitive to EQ and FX skews.
- Capturing skews is critical in order to reflect the cost of hedging the deltas embedded into the quanto covariance
- We proposed an approximation formula for quanto forward and quanto options under local volatilities, which can be used to mark the quanto forward with an FX and EQ skew aware model
- This was achieved by applying a Proxy Expansion technique pioneered by Gobet et al. to calibrate a quadratic local vol model and to derive the approximation formula
- Experience shows that this is a competitive alternative to the full-blown 2-factor forward PDE

Acknowledgement

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References

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